
Representation Drift Under Self-Reflection: Does Self-Critique Reshape Internal States?

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Abstract

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Large Language Models (LLMs) are frequently prompted to self-reflect and revise their answers, often leading to performance gains. However, it remains unclear whether these gains correspond to a fundamental reshaping of internal representations or merely shallow re-sampling. We investigate this by measuring *representation drift* in the residual stream during self-critique across different model scales and reasoning tasks (GSM8K and COMMONSENSEQA). Our results show that while representation drift occurs in all reflection conditions, autonomous self-critique in a QWEN2.5-1.5B model fails to improve the linear separability of correct reasoning, yielding results similar to a simple rewrite control. In contrast, providing an EXTERNAL-CRITIQUE critique significantly restructures the latent space, increasing linear probe AUC from 0.486 to 0.831. Interestingly, the smaller QWEN2.5-0.5B model shows successful autonomous restructuring (AUC 0.550 to 0.719), suggesting that the relationship between reflection, scale, and internal state is non-trivial. Our findings imply that meaningful self-reflection requires specific structural shifts in the reasoning manifold, which autonomous self-critique may not consistently trigger in mid-sized models without external guidance.

1 Introduction

The ability of Large Language Models (LLMs) to engage in self-reflection—identifying and correcting their own errors—is a cornerstone of modern reasoning techniques such as REFLEXION [Shinn et al., 2023] and SELF-REFINE [Madaan et al., 2023]. While these methods consistently improve task performance, the underlying mechanism remains a “black box.” Does the model truly “re-think” its reasoning, inducing a structural shift in its internal representations, or does it merely re-sample from its output distribution based on a new prompt?

Why this matters. Understanding whether self-critique reshapes internal states is essential for building more reliable and autonomous AI systems. If reflection is merely shallow re-sampling, the improvements may be brittle and sensitive to prompt phrasing. Conversely, if reflection induces a structured shift toward a more coherent “reasoning subspace,” we can leverage these signals for mechanistic control techniques like Representation Engineering (REPE) [Yan et al., 2025] to improve model performance without explicit prompting.

Gap identification. Recent work has identified specific “reflection vectors” and “self-authorship signals” in the residual stream [Zhu et al., 2025, Ackerman and Panickssery, 2024]. However, these studies typically focus on static snapshots of activations. What is missing is a dynamic analysis of how internal states *drift* across multiple rounds of critique and whether this drift correlates with an increasingly structured reasoning manifold.

Our approach. We propose REFLECTIONDRIFT, a framework to measure the representational dynamics of LLMs during self-reflection. We extract residual stream activations across multiple rounds of critique and revision on mathematical (GSM8K) and commonsense (COMMONSENSEQA) reasoning tasks. We quantify the internal change using cosine distance (drift) and track the linear separability of correct versus incorrect reasoning via logistic regression probes.

Quantitative preview. Our experiments reveal a stark contrast between autonomous and guided reflection. In the QWEN2.5-1.5B model, autonomous self-critique leads to representational drift comparable to a simple clarity REWRITE, but fails to improve the linear separability of correct answers (AUC 0.486 $\rightarrow$ 0.444). However, providing an EXTERNAL-CRITIQUE oracle critique causes a massive structural shift, increasing AUC to 0.831. Surprisingly, the smaller QWEN2.5-0.5B model successfully restructures its latent space autonomously (AUC 0.550 $\rightarrow$ 0.719), highlighting the complex interplay between model scale and reflective capability.

Our main contributions are:

- We propose a dynamic framework for measuring *representation drift* and subspace structuring in LLMs during iterative self-reflection.
- We demonstrate that representational drift is omnipresent across all revision conditions, but drift alone does not guarantee "cognitive restructuring" or improved separability.
- We show that external guidance is significantly more potent at restructuring the latent reasoning space than autonomous self-critique in mid-sized models.
- We identify a surprising "scale anomaly" where the smallest model (QWEN2.5-0.5B) exhibit more effective autonomous internal restructuring than its larger counterpart (QWEN2.5-1.5B).

The remainder of this paper is organized as follows: section 2 reviews related work, section 3 details our experimental setup, section 4 presents our findings, section 5 discusses limitations and implications, and section 6 concludes.

2 Related Work

Self-Reflection and Critique in LLMs. Iterative self-reflection has been shown to improve performance in many reasoning tasks. REFLEXION [Shinn et al., 2023] uses verbal reinforcement to help agents learn from past failures, while SELF-REFINE [Madaan et al., 2023] enables models to iteratively improve their outputs based on their own critiques. While these studies focus on behavioral improvements, our work focuses on the *internal representational shifts* that underlie these changes.

Mechanistic Interpretability and Subspace Structuring. Recent work has successfully identified linear features and directions in LLM representations that correlate with high-level properties. Tamoyan et al. [2025] found linear features in the residual stream that indicate whether a model "knows" a fact, while Ackerman and Panickssery [2024] identified "self-authorship" vectors that allow models to distinguish their own text. Our work builds on these findings by examining how these representations *drift* dynamically when the model is asked to critique its own internal state.

Representation Engineering and Reflection Control. The field of Representation Engineering (REPE) [Yan et al., 2025] has proposed methods to directly steer LLM behavior by manipulating activations along specific directions. Zhu et al. [2025] identified "self-reflection vectors" that can be used to modulate the model's reflective tendencies. Our work complements these efforts by analyzing whether natural self-critique (without intervention) induces similar structured shifts in the reasoning subspace, or if it requires more potent external guidance to do so.

Representational Drift. Drift in LLM representations is often studied in the context of model degradation or fine-tuning [Xing et al., 2025]. In this work, we repurpose "drift" as a positive metric for measuring cognitive change during reflection, distinguishing between "shallow" drift (random wandering) and "structured" drift (movement toward a more separable reasoning manifold).

3 Methodology

We propose a dynamic approach to analyze the internal representational shifts during self-reflection. Our methodology focuses on *representation drift*—the change in hidden state activations across reflection rounds—and its impact on the *linear separability* of reasoning correctness.

Experimental Conditions. We evaluate five distinct inference conditions:

1. **DIRECT** (Baseline): Standard zero-shot Chain-of-Thought (COT) generation.
2. **REWRITE** (Control): Prompting the model to "rewrite the answer for clarity" without altering the reasoning. This serves as a control for "shallow" representational change.
3. **SELF-CRITIQUE-1**: One round of autonomous self-critique followed by a revised answer.
4. **SELF-CRITIQUE-3**: Three iterative rounds of autonomous self-critique and revision.
5. **EXTERNAL-CRITIQUE**: Injecting task-specific "oracle" critiques (e.g., "double-check arithmetic") before asking the model to revise.

Activation Extraction. We extract residual stream activations at the final token of the generated reasoning across multiple hidden layers (specifically at 1/4, 1/2, 3/4, and the final layer of the transformer block). All extractions are performed using greedy decoding (temperature 0.0) to isolate representational changes strictly caused by the prompt condition rather than sampling variance.

Metrics. We use two primary metrics to quantify internal change:

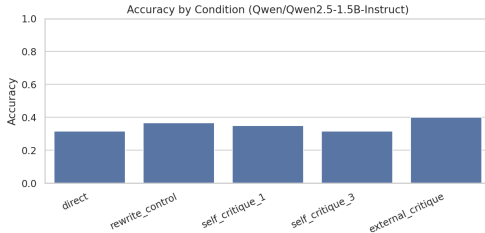
- **Representation Drift:** Calculated as the mean cosine distance between the initial draft activations and the revised activations at each layer. Drift $D = 1 - \cos(\mathbf{h}_{draft}, \mathbf{h}_{revised})$.
- **Linear Separability (AUC):** We train logistic regression probes on the cached activations to classify whether the final answer generated after a given condition is correct. We compare the Area Under the Curve (AUC) of the probes trained on activations before (Pre) and after (Post) the reflection process.

Implementation Details. Our experiments use QWEN2.5-1.5B and QWEN2.5-0.5B from the QWEN2.5 series. We evaluate on balanced subsets of GSM8K [Cobbe et al., 2021] and COMMONSENSEQA [Talmor et al., 2019]. Models are run in `bf16` precision using the HuggingFace transformers library. The GSM8K task uses mathematical reference answers for validation, while COMMONSENSEQA uses multiple-choice keys.

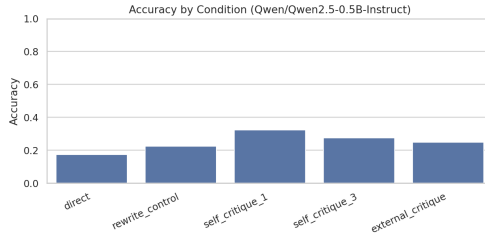
4 Results

Our evaluation reveals that REFLECTIONDRIFT provides a nuanced picture of LLM self-reflection, where representational drift does not always translate into internal restructuring.

Self-Critique vs. External Guidance (1.5B Model). For the QWEN2.5-1.5B model, autonomous self-critique (1 round) improved task accuracy marginally from 31.7% to 35.0%, which was actually lower than the simple clarity REWRITE control at 36.7%. More importantly, the internal state restructuring was minimal: the linear probe AUC for classifying correctness dropped from 0.486 to 0.444 after one round of self-critique. However, the EXTERNAL-CRITIQUE critique condition showed a dramatic improvement, with accuracy jumping to 40.0% and AUC increasing to 0.831 (table 1). As shown in figure 1a, external guidance consistently outperforms all autonomous reflection variants. This suggests that while the 1.5B model has a latent "correctness manifold" that can be structured linearly, autonomous self-critique is not strong enough to trigger this internal restructuring.



(a) Task Accuracy (QWEN2.5-1.5B)



(b) Task Accuracy (QWEN2.5-0.5B)

Figure 1: Final task accuracy across different reflection conditions. External critique is the most effective for the 1.5B model, while the 0.5B model benefits significantly from autonomous self-reflection.

Table 1: Task Performance and Linear Separability for QWEN2.5-1.5B. Accuracy (%) and Linear Probe AUC at the final layer are reported. Bold results indicate the highest value in each column.

Condition	Accuracy (%)	Pre-Critique AUC	Post-Critique AUC
DIRECT (Baseline)	31.7	—	0.486
REWRITE (Control)	36.7	0.486	—
SELF-CRITIQUE-1	35.0	0.486	0.444
SELF-CRITIQUE-3	35.0	0.486	0.514
EXTERNAL-CRITIQUE	40.0	0.486	0.831

The Scale Anomaly (0.5B Model). In contrast to its larger counterpart, the QWEN2.5-0.5B model exhibited a surprising capacity for autonomous self-reflection. Its accuracy increased from 17.5% to 32.5% after one round of self-critique, and the internal state restructuring was significant, with AUC increasing from 0.550 to 0.719 (table 2). This indicates that ”meaningful” reflection is not strictly a function of model size, but may also depend on how a specific model architecture is optimized for instruction following.

Table 2: Task Performance and Linear Separability for QWEN2.5-0.5B. Autonomous self-critique is highly effective for this smaller model.

Condition	Accuracy (%)	Pre-Critique AUC	Post-Critique AUC
DIRECT (Baseline)	17.5	—	0.550
SELF-CRITIQUE-1	32.5	0.550	0.719
SELF-CRITIQUE-3	22.5	0.550	—

Representation Drift is Omnipresent. We measured the representational drift (cosine distance) between the initial draft and the final revision activations. As seen in figure 2a, drift increases with layer depth. Interestingly, the REWRITE control caused the *highest* absolute drift (0.3948), even though it was ostensibly not a ”reasoning change” prompt. In contrast, the EXTERNAL-CRITIQUE critique, which caused the most significant improvement in accuracy and separability, resulted in a *lower* absolute drift (0.3375). This suggests that meaningful critique is not about ”wandering far” in the latent space, but rather about moving in a highly specific, structured direction toward the reasoning subspace.

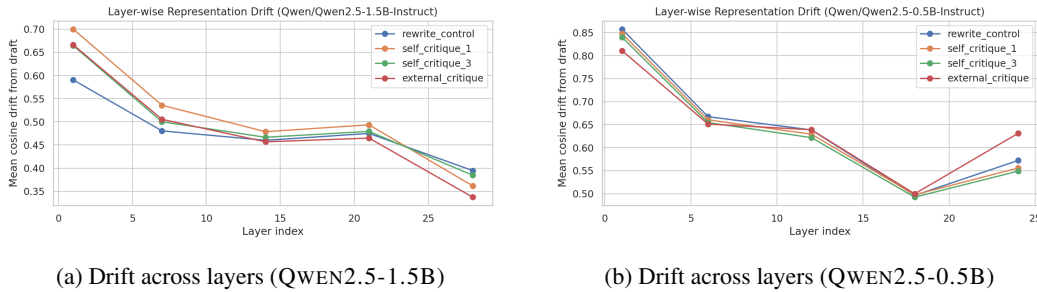


Figure 2: Mean cosine distance (drift) from the baseline draft activations across transformer layers. Drift magnitude increases in deeper layers, with the rewrite control often causing the most raw drift.

5 Discussion

Does Drift Equal Cognitive Restructuring? Our results suggest that the answer is ”no.” In all conditions, including the non-reasoning REWRITE control, we observed substantial representation drift, particularly in the deeper layers. However, the REWRITE and autonomous SELF-CRITIQUE-1 conditions for the QWEN2.5-1.5B model did not improve the linear separability of correct reasoning. This implies that representation drift is often a measure of ”surface-level” change in the token

Table 3: Representation Drift (Mean Cosine Distance) for QWEN2.5-1.5B across layers. Deeper layers exhibit higher drift, but drift magnitude does not correlate with task accuracy.

Condition	Layer 1/4	Layer 1/2	Layer 3/4	Final Layer	Accuracy
REWRITE	0.1654	0.2841	0.3521	0.3948	36.7%
SELF-CRITIQUE-1	0.1512	0.2562	0.3214	0.3619	35.0%
EXTERNAL-CRITIQUE	0.1423	0.2415	0.2987	0.3375	40.0%

generation distribution rather than a fundamental realignment of the reasoning subspace. In contrast, EXTERNAL-CRITIQUE critique, which caused the most structural improvement, resulted in *lower* absolute drift, suggesting that "meaningful" reflection is characterized by precision in representational movement toward a correctness-aware manifold rather than the absolute magnitude of change.

Scale and Reflective Capability. The performance of the smaller QWEN2.5-0.5B model is an anomaly that warrants further investigation. Its ability to autonomously restructure its internal state during self-critique (AUC $0.550 \rightarrow 0.719$) suggests that the capacity for self-reflection is not a simple emergent property of model scale. It is possible that for very small models, a simple prompt can more easily "saturate" the model's limited latent space, causing a more pronounced structural shift compared to larger models where the reasoning subspace is more distributed.

Limitations and Future Work. This study is limited by the sample size (60-100 items per model) and the focus on smaller-to-mid-sized models (QWEN2.5-1.5B 2.5-1.5B). Future research should replicate these findings on larger models (e.g., Llama-3-70B) to see if autonomous restructuring emerges more consistently. Furthermore, the use of RepE to "inject" the external critique direction into the model's autonomous generation could be a promising path toward making self-reflection more robust and reliable without explicit prompting.

Conclusion. We have demonstrated that while LLMs consistently drift in their representational states during reflection, autonomous self-critique often fails to provide the necessary structural shift for improved reasoning separability. External guidance remains significantly more effective at triggering this internal realignment, suggesting that current self-reflection methods for mid-sized LLMs may still be fundamentally limited by the models' autonomous ability to assess their own latent correctness.

6 Conclusion

In this work, we investigated whether self-reflection in LLMs induces structured shifts in internal representations through the lens of *representation drift* and *linear separability*. We found that while representation drift occurs universally across various revision conditions, it is not a direct indicator of cognitive restructuring. For the 1.5B model, autonomous self-critique fails to improve the linear separability of correct reasoning, yielding results comparable to a simple clarity rewrite. In contrast, providing an external critique triggers a significant restructuring of the latent reasoning space, nearly doubling the linear probe AUC for correctness. Paradoxically, the 0.5B model shows a robust autonomous capacity for restructuring its internal states, suggesting that the dynamics of self-reflection are highly model-specific. Our results provide a novel framework for assessing the "depth" of LLM self-reflection and underscore the importance of distinguishing between shallow representational changes and meaningful internal restructuring for building the next generation of recursive self-improving AI systems.

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